# Group Theory

Definition 4.1 (Group). A group is a set G with a binary operation * such that:

(i) Closure: a*b in G for all a,b in G.

(ii) Associativity: (a*b)*c = a*(b*c) for all a,b,c in G.

(iii) Identity: there exists e in G with e*a = a*e = a for all a in G.

(iv) Inverse: for each a in G there exists a' with a*a'=a'*a=e.

Definition 4.2 (Abelian Group). A group is abelian if a*b = b*a for all a,b.

Example 4.3. The integers Z under addition form an abelian group.

Definition 4.4 (Subgroup). A subset H of a group G is a subgroup if H is itself a group under the operation of G.

Theorem 4.5 (Lagrange's Theorem). If G is a finite group and H is a subgroup of G, then the order of H divides the order of G.